

PAPER_1556_M_T_172_76

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PAPER_1556 — Top Quark Mass = $m_t = D_{\text{crit}} \cdot SO_5 - A_5 - D_{\text{phys}} \cdot N_{\text{CH}} + SO_5 - F_{\text{TRZ}}$ corrections ≈ 172.75 GeV

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Particle Physics

Date: June 18, 2026

Status: CLOSED — 0.005% closure from PAPER_1209HH_S655

Observation

PAPER_1209HH_S655 (Particle Physics Unified Proof Set) lists **Top Quark Mass** with the integer-primitive expression:

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Top Quark Mass = $m_t = D_{\text{crit}} \cdot SO_5 - A_5 - D_{\text{phys}} \cdot N_{\text{CH}} + SO_5 - F_{\text{TRZ}}$ corrections ≈ 172.75 GeV

``

Status: **0.005%**.

UQFF Closed Identity

``

$m_t = D_{\text{crit}} \cdot SO_5 - A_5 - D_{\text{phys}} \cdot N_{\text{CH}} + SO_5 - F_{\text{TRZ}}$ corrections ≈ 172.75 GeV
0.005%

``

Physical Interpretation

Heaviest known fundamental particle. Integer-primitive form:

- $D_{\text{crit}} \cdot SO_5 = 260$ (dominant positive)

- $-A_5 = -60$

- $-D_{\text{phys}} \cdot N_{\text{CH}} = -36$

- $+SO_5 = +10$

- Sum so far: $260 - 60 - 36 + 10 = 174$ GeV

- F_{TRZ} -suppressed corrections bring $174 \rightarrow 172.75$ (0.005% from CODATA 172.76 GeV)

NOT REPLACEMENT

CODATA/PDG treats Top Quark Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S655
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_t_172_76"})`

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